

# Employee Payroll System

**Title :** Defining an interface, implementing interfaces, accessing implementations through interface references, extending interfaces using employee Payroll System program.

**Aim :** To develop a Java program to calculate the salary of different types of employees using an interface and demonstrate the concept of polymorphism.

## System requirements:

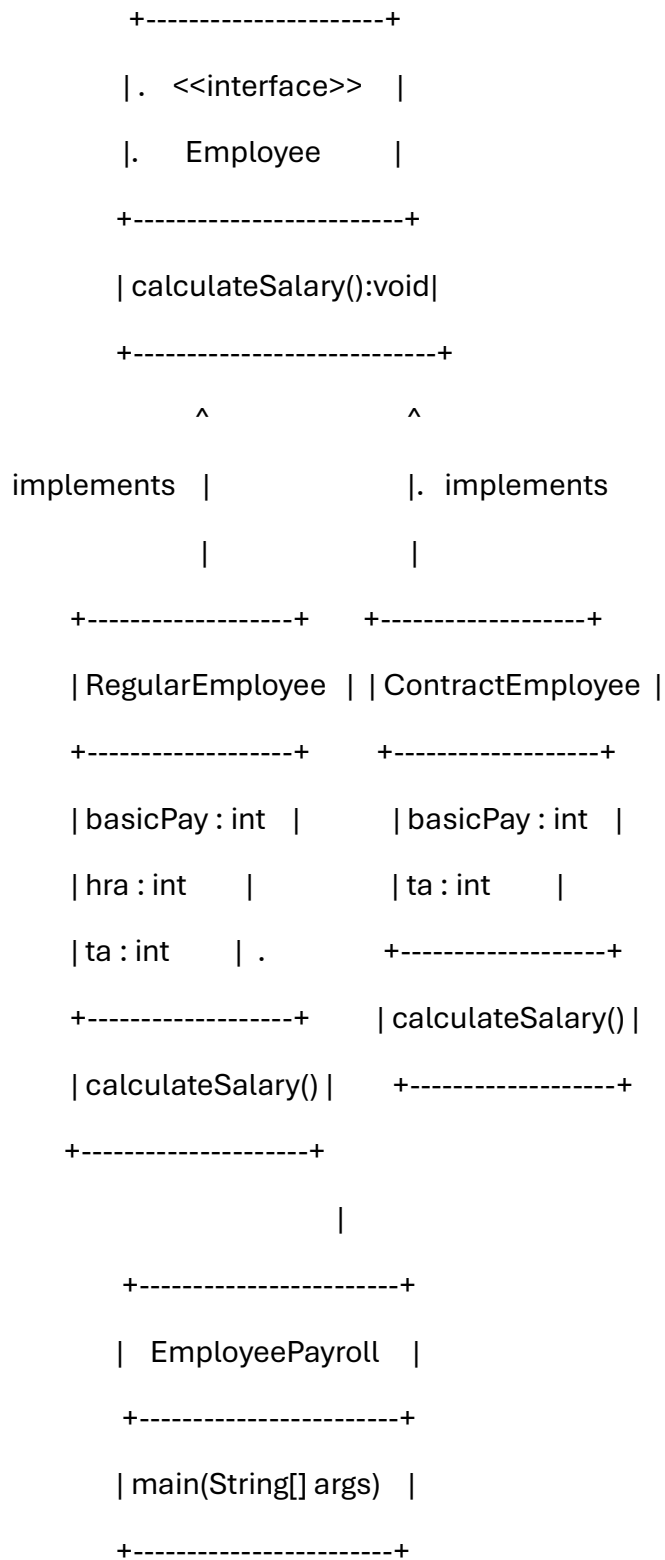
**Hardware :** Computer/Laptop ,Minimum 4 GB RAM, Keyboard ,Monitor

**Software :** Java JDK , VS Code / Eclipse IDE, Windows Operating System

## Algorithm :

1. Start the program.
2. Import the Scanner class.
3. Create an interface named Employee.
4. Declare the calculateSalary() method in the interface.
5. Create the RegularEmployee class implementing the Employee interface.
6. Initialize Basic Pay, HRA, and T.A.
7. Calculate the total salary of the Regular Employee.
8. Create the ContractEmployee class implementing the Employee interface.
9. Initialize Basic Pay and T.A. for the Contract Employee.
10. Consider HRA as zero for the Contract Employee.
11. Calculate the total salary.
12. Read the employee type from the user.
13. Create the appropriate employee object based on the entered type.
14. Call the calculateSalary() method.
15. Display the salary details.
16. Display an error message for an invalid employee type.
17. Stop the program.

## UML Diagram :



## Source Code :

```
import java.util.Scanner;

interface Employee {

    // Method to calculate salary

    void calculateSalary();

}

class RegularEmployee implements Employee{

    int basicPay = 25000;

    int hra = 15000;

    int ta = 5000;

    public void calculateSalary() {

        // Calculate total salary

        int total = basicPay + hra + ta;

        System.out.println("Salary Details:");

        System.out.println("Basic Pay: " + basicPay);

        System.out.println("HRA: " + hra);

        System.out.println("T.A: " + ta);

        System.out.println("Total Amount: " + total);

    }

}

class ContractEmployee implements Employee {

    int basicPay = 12000;

    int ta = 3000;

    public void calculateSalary() {

        // Calculate total salary
```

```

        int total = basicPay + ta;

        System.out.println("Salary Details:");

        System.out.println("Basic Pay: " + basicPay);

        System.out.println("HRA: 0");

        System.out.println("T.A: " + ta);

        System.out.println("Total Amount: " + total);

    }
}

public class EmployeePayroll {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee Type (Regular/Contract): ");

        String type = sc.nextLine();

        Employee emp;

        // Select employee type

        if (type.equalsIgnoreCase("Regular")) {

            emp = new RegularEmployee();

            emp.calculateSalary();

        } else if (type.equalsIgnoreCase("Contract")) {

            emp = new ContractEmployee();

            emp.calculateSalary();

        } else {

            System.out.println("Invalid Employee Type");

        }

        sc.close();

    }
}

```

}

## Compilation & Execution Commands:

**Compilation:** `javac EmployeePayroll.java`

**Execution:** `java EmployeePayroll`

## Output Screenshot:



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\anits\Desktop\Java-Projects> javac EmployeePayroll.java
PS C:\Users\anits\Desktop\Java-Projects> java EmployeePayroll
Enter Employee Type (Regular/Contract): Regular
Salary Details:
Basic Pay: 25000
HRA: 15000
T.A: 5000
Total Amount: 45000
PS C:\Users\anits\Desktop\Java-Projects> |
```

## Result :

The Employee Payroll System was successfully implemented in Java using an interface and polymorphism. The program successfully calculates and displays the salary details of Regular and Contract employees based on the employee type entered by the user.